5 (a)	Magnetic flux density is defined as the force acting per unit current in a wire of unit length placed at right-angles to the field {accept: equation $B = F/ILsin\theta$, with all symbols clearly defined}	[1]
(b)(i)	$F = Bqv \sin 90^\circ$ $= (20\times10^{-3})(3\times10^{-6})(1.5/100)$ $= 9.0 \times 10^{-10}$ N	 [1] [1]
(ii)	(Using Right-hand grip rule) direction of B-field is into paper at the point charge (due to wire), and using Fleming's Left Hand Rule, direction of magnetic force is to the right, hence the charge will therefore be deflected towards the right as it moves ahead.	[1] [1]
(c)(i)	Using Right-hand grip rule, B-field due to long wire at P is into plane. Using Right-hand grip rule, B-field due to coil at P is out of plane. {accept: both fields in "opposite directions"}	[1]

	Hence resultant field is <u>smaller</u> when there is a current flowing in the coil.	[1]
(ii)	Coil will be <u>pushed away</u>. (Since the part closer to the conductor will be repelled, unlike directions of current)	[1]
(d)(i)	<u>Parallel</u> (to the field)	[1]
(ii)	(max) torque = F × d $2.1 \times 10^{-3} = F \times 2.8 \times 10^{-2}$ F = 0.075 N (use of 4.5 cm scores no marks)	 [1] [1]
(iii)	$F = NBIL \sin\theta$ $0.075 = 140 \times B \times 0.170 \times 4.5 \times 10^{-2} \times \sin 90^{o}$ $B = 7.0 \times 10^{-2}$ T	 [1] [1]
(iv)	change in flux linkage = NBA $= 140 \times 0.070 \times (4.5 \times 10^{-2} \times 2.8 \times 10^{-2})$ = 0.0123 Wb average induced e.m.f. = 0.0123 / 0.14 = 0.088 V	[1] [1]